

SENSORY NERVOUS SYSTEM

1. Regarding synapses:

- a- presynaptic knobs contain vesicles which have t-snare in their membranes
- b- v-snare of the vesicles has high affinity to t-snare in the active zone of the presynaptic membrane .
- c- Ca^{2+} voltage gated channels of the pre-synaptic knob open during repolarization
- d- synaptic cleft contain high concentration of K^+ .

2. Concerning organization of synapses:

- a- axo-axonic is the most excitable.
- b- axo-dendritic is the most excitable
- c- axo-somatic is the least excitable.
- d- axo-dendritic is the least in number.

3. Regarding the mechanism of synaptic transmission:

- a- Ca^{2+} ions are released from the presynaptic knob.
- b- presynaptic knobs release only an excitatory transmitter.
- c- chemical transmitter causes development of only excitatory post-synaptic potentials.
- d- excitatory transmitters open Na^+ channels.

4. Synaptic transmission is characterized by all of the following, EXCEPT:

- a- impulses are transmitted from the pre to the postsynaptic neurons
- b- alkalosis depresses synaptic transmission.
- c- fatigue is due to exhaustion of neurotransmitter stores.
- d- repetitive stimulation leads to synaptic potentiation.

5. Released transmitter is inactivated by all of the following except:

- a- active re-uptake.
- b- activation by enzymes.
- c- diffuse away in ECF.
- d- phagocytosed by glial cells.

6. Regarding synaptic potentials:

- a. presynaptic inhibition causes hyperpolarization of post-synaptic neuron.
- b. during IPSP a neuron can not discharge an action potential.
- c. presynaptic facilitation is due to increased secretion of chemical transmitter due to stimulation by a 3rd excitatory presynaptic neuron.
- d. EPSP can be caused by glycine.

7. Excitatory postsynaptic potential:

- a. can not be summated.
- b. is a state of local depolarization.
- c. is due to opening of voltage gated Na^+ channels.
- d. is due to pening of chemical gated K^+ channels.

8. Excitatory postsynaptic potential:

- a- is due to increase permeability of the postsynaptic membrane to Na^+
- b- is a state of complete depolarization.
- c- can produce an action potential without summation.
- d- of dendrites can reach to an action potential.

9. EPSPs :

- a- result from release of GABA from presynaptic neuron.
- b- can be temporally summated.
- c- result from increase permeability of the postsynaptic membrane to K^+
- d- obeys the all or none rule.

10. Excitatory postsynaptic potential:

- a- is not propagated.
- b- shows reversal of polarity.
- c- lasts about 2 seconds.
- d- may reach action potential at the soma.

11. An inhibitory neurotransmitter increases the permeability of the postsynaptic neurone to:

- a- K^+
- b- Na^+
- c- Ca^{2+}
- d- HCO_3^-

12. Inhibitory post synaptic potentials:

- a- is due to presynaptic inhibition.
- b- causes a state of hyperpolarization.
- c- is associated with influx of Na^+ ions.
- d- can not be summated.

13. IPSP:

- a- is a state of depolarization of the postsynaptic membrane.
- b- is produced by binding of acetylcholine to its receptor.
- c- results from inflow of Cl^- ions and outflow of K^+ ions from the postsynaptic membrane.
- d- results from the release of catecholamines from the presynaptic knob.

14.IPSPs:

- a- are a consequence of presynaptic inhibition.
- b- interact with other fast and slow potentials to move the membrane potential of the postsynaptic neuron toward or away
- c- can be produced by an increase in Na^+ conductance from the firing level.
- d- can be produce by an increase in Ca^{2+} conductance.

15.Inhibitory post synaptic potential:

- a- involves increased K^+ efflux.
- b- involves increased Na^+ efflux.
- c- is a state of partial depolarization.
- d- increases excitability of the postsynaptic membrane.

16.Regarding presynaptic inhibition:

- a- the postsynaptic neuron membrane is hyperpolarized.
- b- there is decrease release of excitatory chemical transmitters from the presynaptic neuron.
- c- there is increased Ca^{2+} influx into presynaptic neuron.
- d- there is closure of K^+ channels of the presynaptic neuron.

17. Inhibitory presynaptic potential:

- a- needs inhibitory third neuron.
- b- needs excitatory third neuron.
- c- leads to depolarization of the presynaptic neurone.
- d- directly inhibits the postsynaptic neurone,

18.Grand post-synaptic potential, may cause the postsynaptic membrane to:

- a- be more excitable if $\text{EPSP} > \text{IPSP}$.
- b- be less excitable if $\text{IPSP} > \text{EPSP}$.
- c- reach firing if there is summation of EPSPs
- d- all of the above

19.Regarding synaptic transmission:

- a- is depressed by alkalosis.
- b- is increased by acidosis.
- c- repetitive stimulation of a single presynaptic neurone leads to development of an action potential in the postsynaptic neurone
- d- is increased by hypoxia.

20. Synaptic transmission:

- a- is increased by alkalosis.
- b- shows a delay of about 0.05 ms.
- c- is increased by acidosis.
- d- decreased during tetanus disease

21. Synaptic transmission is facilitated by all of the followings, EXCEPT:

- a- coffee and tea.
- b- alkalosis.
- c- strychnine.
- d- hypoxia.

22. Regarding synaptic transmission, it is decreased by:

- a- alkalosis.
- b- hyperventilation.
- c- increased secretion of GABA
- d- theobromine.

23. If the central delay in one reflex is 4 msec., then the:

- a- number of synapses in this reflex is two.
- b- number of synapses in this reflex is four.
- c- number of synapses in this reflex is eight.
- d- it is a monosynaptic reflex.

24. Synaptic plasticity shows all of the following, EXCEPT:

- a- ability to change the function of the synapse according to the demand.
- b- play an important role in memory.
- c- may lead only to long term potentiation.
- d- explain the process of habituation.

25. Sensitization of synapses:

- a- is due to repetitive benign stimuli which is associated with nocuous stimuli.
- b- is the base of long term memory.
- c- is explained by postsynaptic facilitation.
- d- causes hyperpolarization of the postsynaptic neurone.

26. Action potential is:

- a- initiated at dendrites.
- b- initiated at soma.
- c- initiated at axon hillock.
- d- due to summations of IPSPs.

27. Glycine:

- a- acts mainly as inhibitory chemical transmitter.
- b- action is potentiated by strychnine.
- c- causes depolarization of the postsynaptic membrane.
- d- is responsible for the process of sensitization

28.Regarding rapidly acting transmitters , they are :

- a- formed in the soma.
- b- formed in small amounts.
- c- very potent.
- d- responsible for rapid neurological response.

29.Concerning the slowly acting neurotransmitters , they are :

- a- formed in the synaptic knob.
- b- formed in huge amounts.
- c- stored in clear vesicles in synaptic knobs
- d- very potent

30. Regarding gamma-aminobutyrate (GABA) secreted by a 3rd inhibitory pre-synaptic neurone , all of the following statements are true, EXCEPT:

- a- it is an excitatory neurotransmitter in the brain.
- b- it produces an IPSP.
- c- it decreases Ca^{2+} influx into the pre-synaptic neurone.
- d- it increases Cl^- influx and K^+ efflux in the post-synaptic neurone.

31. NMDA receptors:

- a- are stimulated by acetyl choline .
- b- are ligand-gated cation channels.
- c- are blocked by Ca^{2+} at normal membrane potential
- d- are inhibited by consequent activation of AMPA receptors.

32. Divergence:

- a- helps amplification of the signal.
- b- means that several input fibres supply a single neurone.
- c- helps the process of summation.
- d- explains the process of occlusion.

33. The following are true regarding convergence, EXCEPT:

- a- many input signals supply one neurone.
- b- leads to summation.
- c- explain the process of occlusion.
- d- helps passage of a signal into many pathways

34. Facilitation:

- a- is due to stimulation of 2 input neurons away from each other.
- b- is due to stimulation of 2 input neurons close to each other.
- c- number of action potentials released is less than if each input neurone is stimulated alone.
- d- occurs when the two afferent neurons have a common discharge zone.

35. During facilitation:

- a- the two input afferent fibres diverge.
- b- the two input afferents are close to each other.
- c- there is overlap of the facilitated zones.
- d- there is overlap of the discharge zones.

36. Occlusion:

- a- due to stimulation of two afferent neurons away from each other.
- b- occurs when two afferent neurons shared a common discharge zone.
- c- results in greater response than if each afferent neurone is stimulated alone.
- d- occurs when 2 afferent neurons shared a common facilitated zone.

37. Lateral inhibition:

- a- needs only inhibitory neurons.
- b- helps sharpening of sensation.
- c- means that inhibited central neurons are surrounded by excitatory neurons.
- d- needs only one type of chemical transmitter.

38. Renshaw cell:

- a- shows the phenomenon of lateral inhibition.
- b- is activated by glycine.
- c- is present in dorsal horn of the spinal cord.
- d- are excitatory neurons.

39. Pacinian corpuscles are:

- a- pain receptors.
- b- rapidly adapting touch receptors.
- c- slowly adapting touch receptors.
- d- innervated by C nerve fibers.

40. The receptor potential:

- a- is state of hyperpolarization in the receptor membrane.
- b- is caused by specific increase in the permeability of the receptor membrane to Na⁺.
- c- has a duration more than 2 msec.
- d- has an amplitude inversely proportional to the intensity of the stimulus.

41. Paccinian corpuscles are:

- a- type of temperature receptors.
- b- usually innervated by A δ nerve fibre.
- c- rapidly adapting receptors.
- d- pain receptors.

42. Specificity:

- a- explains the process of adaptation.
- b- means that each receptor is most sensitive to a specific form of energy called the adequate stimulus.
- c- is a state of partial depolarization.
- d- is due to high intensity of the stimulus.

43. Generator potential:

- a- is a state of partial depolarization.
- b- duration is shorter than that of the action potential.
- c- is accompanied with absolute refractory period.
- d- can not be summated.

44. Receptor potential:

- a- is a state of hyperpolarization.
- b- is conducted passively by the process of electronic depolarization.
- c- generates only one action potential.
- d- is due to specific opening of K⁺ channels.

45. Receptor potential:

- a- is an action potential.
- b- is due to non-specific opening of Na⁺ channels.
- c- duration is shorter than that of the action potential.
- d- brings the membrane potential away from the firing level.

46. Receptor potential:

- a- is actively conducted to the 1st node of Ranvier.
- b- has a duration shorter than that of the action potential.
- c- amplitude is proportional to the intensity of the stimulus.
- d- is due to specific increase of the membrane permeability to Na⁺ ions

47. The receptor potential:

- a- is conducted actively to the 1st node of Ranvier.
- b- can be summated.
- c- is accompanied with absolute refractory period.
- d- has the same duration like that of the action potential.

48. Fast adaptation receptors include:

- a- pain receptors.
- b- auditory receptors.
- c- paccinian corpuscles.
- d- arterial baroreceptors.

49. Discrimination of the intensity of stimuli results from the:

- a- presence of adequate stimulus for each receptor.
- b- presence of specific pathway for each sensation.
- c- specific centre for each sensation.
- d- Weber-Fechner law or Power law.

50. The size of representation of various body parts in the primary somatic sensory cortex is dependent on:

- a- relative size of the body parts .
- b- density of the receptors in that organ.
- c- size of the muscles in that body part.
- d- conduction velocity of the afferent fibers

51. Kinesthetic sensations:

- a- have their receptors in the joint capsule.
- b- allow weight discrimination.
- c- are conducted via ventral spinothalamic tract.
- d- is the perception rapid repetitive stimuli.

52. Fine touch sensations:

- a- are not sharply localized.
- b- need a relatively high threshold stimulus.
- c- are transmitted via dorsal column.
- d- are perceived by free nerve endings.

53. Fine touch sensation:

- a- is perceived by Merkel's discs.
- b- are conducted by C fibres.
- c- is finally relayed in the thalamus.
- d- is perceived more on the back of the shoulder.

54. Two-point discrimination is more accurate:

- a- at the back.
- b- the more the convergence of afferents.
- c- the more the number of receptors.
- d- the less the cortical representation.

55. Action potentials recorded after stimulation of thermal receptors show that:

- a- the maximal rate of discharge of action potentials in the cold nerve fibres is at 25°C.
- b- action potentials are recorded in warm nerve fibres between 10-35°C.
- c- action potentials are recorded in cold pain fibres at 0°C.
- d- the maximum rate of discharge from warm pain and cold pain fibres occurs at 40°C.

56. The distance by which two touch stimuli must be separated to be perceived as two separate stimuli is greatest on:

- a- the lips.
- b- the palm of the hand.
- c- the back of the scapula.
- d- the tips of the finger.

57. Vibration sense:

- a- is sensed by free nerve endings.
- b- depends on past cortical experience.
- c- is carried on gracil and cuneate tracts.
- d- is carried on spinothalamic tract.

58. The fast pain sensation that accompanies a painful trauma is carried to C.N.S. by fibres that belong to:

- a- A alpha fibres.
- b- A delta fibres.
- c- C fibres.
- d- B fibres.

59. Pain:

- a- receptors are stimulated by a rise in bradykinin concentration.
- b- threshold differs from one person to another.
- c- sensation is enhanced by endorphins.
- d- sensation is enhanced by GABA released by A β fibres.

60. The pricking pain:

- a- terminate in the non-specific thalamic nuclei.
- b- is lost by compression of the nerve.
- c- is blocked by local anesthetics better than slow pain.
- d- is a severe dull aching pain.

61. Slow pain:

- a- is carried along type A delta fibres.
- b- pathway activates the non-specific thalamic nuclei.
- c- is perceived mainly in sensory area I.
- d- lasts for a short time.

62. Cutaneous pain:

- a- receptors adapt to stimulation more rapidly than touch receptors.
- b- can be caused by excitation of receptors by K^+ ions released in injured tissues.
- c- can be caused by over-stimulation of touch receptors.
- d- is carried into the spinal cord by a long A beta fibers.

63. Cutaneous pain :

- a- threshold is the same for males and females.
- b- is carried only on myelinated nerve fibres.
- c- is carried on the spinothalamic tract of the same side.
- d- may refer to the viscera

64. Deep pain:

- a- is due to stimulation of pain receptors in the skin.
- b- is pricking in character.
- c- is poorly localized.
- d- is carried along a delta fibres.

65. Concerning visceral pain, the followings are true statements, EXCEPT:

- a- may cause reflex vomiting.
- b- may cause contraction of overlying skeletal muscle.
- c- is well localized.
- d- may cause change in arterial blood pressure.

66. All the following may give rise to visceral pain, EXCEPT:

- a- distension of hollow viscus.
- b- ischemia.
- c- spasm of hollow viscus.
- d- incision with scalpel in the gut wall.

67. Visceral pain:

- a- is a localized pain.
- b- is accompanied by spasm of its overlying muscle.
- c- can be elicited by sharp cutting of viscera.
- d- is carried along garcail and cunate tracts.

- 68. Visceral pain results from all of the following, EXCEPT:**
- a- ischaemia which releases bradycardia.
 - b- Escape of HCl through rupture duodenal ulcer.
 - c- overdistension of hollow organ.
 - d- cutting through the viscera.
- 69. Gallbladder pain is referred to the :**
- a- umbilicus.
 - b- tip of the right scapula.
 - c- left shoulder.
 - d- left lumbar region.
- 70. The analgesia system includes:**
- a- Raphe magnus nucleus in the post horn of the spinal cord.
 - b- the periaquiductal area in the mid-brain.
 - c- the sensory area I.
 - d- supra-optic nuclei of the hypothalamus
- 71. Regarding the analgesia system:**
- a- rubbing of the skin inhibits slow pain by a presynaptic inhibition mechanism.
 - b- periaqueductal neurons secrete enkphalin.
 - c- periaqueductal neurons are stimulated by serotonin.
 - d- interneurons of the dorsal horn causes postsynaptic inhibition of the Raphe magnus nucleus.
- 72. Serotonin is used as neurotransmitter in pain suppression by neuron in the following region :**
- a- raphe magnus nucleus .
 - b- periaqueductual gray matter .
 - c- post-central gyrus .
 - d- nucleus cuneatus .
- 73. An interneuron in which region uses enkephalin to inhibit pain transmission ?**
- a- post-central gyrus .
 - b- raphe magnus nucleus .
 - c- dorsal horn of spinal cord .
 - d- periaqueductal gray matter .

- 74. An interneuron in which region uses enkephalin to inhibit pain transmission ?**
- a- post-central gyrus .
 - b- raphe magnus nucleus .
 - c- dorsal horn of spinal cord .
 - d- periaqueductal gray matter .
- 75. Tactile stimulation of skin surface for pain inhibition involves the following fibers:**
- a- A δ fibers in peripheral nerves.
 - b- C fibers in peripheral nerves.
 - c- A β fibers in peripheral nerves.
 - d- A α fibers in peripheral nerves.
- 76. Headache can be produced by all of the following, EXCEPT:**
- a- irritation of meninges.
 - b- dilatation of intracranial blood vessels.
 - c- constriction of extracranial blood vessels.
 - d- loss of cerebrospinal fluid
- 77. Intracranial headache:**
- a. is a type of referred pain.
 - b. may be caused by error of refraction.
 - c. can be caused by lumbar puncture.
 - d. can be produced by constriction of intracranial blood vessels.
- 78. Intracranial headache may be caused by :**
- a- errors of refraction
 - b- spasm neck muscles
 - c- lumbar puncture
 - d- sinusitis.
- 79. Prodromal symptoms experienced by patients with migraine headache is caused by:**
- a- an increased blood flow to the brain tissue .
 - b- vasospasm of cerebral vessels .
 - c- decreased GABA secretion in the sensory areas of the cortex .
 - d- excessive sleep and inactivity .

- 80. The spinothalamic system transmits the following sensations:**
- a- kinesthetic sensations.
 - b- vibration sensation.
 - c- pain and temperature.
 - d- two-point discrimination
- 81. The anterolateral system in the spinal cord:**
- a- has final relay in the gracil and cuneate nuclei.
 - b- carries pain from the same side of the body.
 - c- decussate at the level of the medulla.
 - d- decussate in front of the central canal of the spinal cord.
- 82. The main final destination of signals that carry kinesthetic sensations is the:**
- a- post-central gyrus of the cerebral cortex.
 - b- ventrolateral nuclei of the thalamus.
 - c- cerebellar cortex.
 - d- caudate nucleus.
- 83. Sexual sensations ascend to the brain through:**
- a- spinocerebellar tracts
 - b- spinothalamic tracts.
 - c- dorsal column of spinal cord.
 - d- reticulospinal tract
- 84. Lesion of the dorsal column may lead to:**
- a- loss of pain and temperature on same side of the lesion.
 - b- positive Romberge's sign.
 - c- loss of vibration sense on the opposite side.
 - d- dissociated sensory loss of touch sensation.
- 85. Tabes dorsalis may lead to:**
- a- loss of sense of position.
 - b- high steppage gate.
 - c- sensory ataxia
 - d- all of the above
- 86. Regarding secondary hyperalgesia:**
- a- pain threshold is low.
 - b- may be due to sun burn.
 - c- skin is reddish in colour.
 - d- may be caused by thalamic syndrome.

87. Concerning primary hyperalgesia:

- a- pain threshold is high.
- b- may be due to sun burn.
- c- skin is normal.
- d- may be caused by thalamic syndrome.

88. A subject suffers from right hemisection of the spinal cord shows:

- a- upper motor neurone paralysis at the level of the hemisection.
- b- loss of pain and temperature on the right side below the level of the section.
- c- loss of fine touch and vibration sense on the left side below the level of the section.
- d- loss of all sensations on the same side at the level of the section

89. Posterior column damage of the spinal cord may impair the following, EXCEPT:

- a- the ability to stand steadily with the eyes closed.
- b- pain sensation.
- c- vibration sensation.
- d- kinesthetic sensations.

90. A patient who has had destruction of somatic sensory area I on the left side will show:

- a- total lack of pain sensation on the same side:
- b- total lack of temperature sensation of the same side.
- c- Inability to identify objects put in the left hand.
- d- inhibitory to identify objects put in the right hand.

Motor Nervous System

(1) The stretch reflex:

- a) is a polysynaptic reflex.
- b) is a monosynaptic reflex.
- c) receptor is the paccianian corpuscle.
- d) has a long central delay.

(2) Regarding the muscle spindle, all are true, EXCEPT:

- a) the primary afferent fibres responds to the static length of muscle as well as to its rate of change in length.
- b) stretching the extrafusal fibres causes the intrafusal muscle fibres to be stretched.
- c) they are excited during overstretch of the muscle .
- d) they are in series with the extrafusal muscle fibres.

(3) All are true regarding the stretch reflex, EXCEPT:

- a) can be elicited by passive stretch.
- b) depends on the muscle spindle which is in parallel with the extrafusal muscle fibres.
- c) it is a polysynaptic reflex.
- d) stretch of a muscle is followed by its contraction .

(4) The gamma efferent system:

- a) controls spindle activity during voluntary movements.
- b) supplies the central zone of intrafusal muscle fibres.
- c) is inhibited by the vestibule-spinal tract.
- d) when stimulated, causes less discharge of muscle spindles.

(5) The primary endings innervating the muscle spindle:

- a) only innervates the nuclear chain fibres.
- b) innervate both nuclear chain and nuclear bag fibres.
- c) are a slowly conducting nerves having a diameter of 4 microns.
- d) discharge during the static response only.

(6) The nuclear bag intrafusal fibers that signal the rate of changes in muscle length is innervated by:

- a) corticospinal neurons .
- b) alpha motor neurons .
- c) dynamic gamma motor neurons .
- d) static gamma motor neurons.

(7) Gamma motor nerve fibers :

- a) cause the contraction of the peripheral part of the intrafusal fibers.
- b) cause contraction of the middle portions of the muscle spindle.
- c) cause relaxation of the intrafusal fibres.
- d) Thickness is about 16 μ

(8) Degeneration of dopamine-containing neurons in the substantia nigra produces:

- a) Choreiform movements .
- b) hemiballismus .
- c) suppression of activity in the subthalamic nucleus .
- d) parkinsonism .

(9) Muscle tone, all are true, EXCEPT:

- a) is defined as resistance offered by muscle to passive stretch.
- b) is maintained by gamma efferent discharges.
- c) maintains body posture against effect of gravity.
- d) is decreased in UMNL .

(10) Regarding the muscle tone, all are true, EXCEPT:

- a) the primary and secondary afferents are both stimulated when the spindle is stretched.
- b) the gamma efferents discharge are firing independently of the descending tracts from areas in the brain.
- c) hypertonism may occur during hyperthyroidism.
- d) hypotonia commonly occurs as a result of spinal motor neuron damage.

(11) Hyporeflexia is present in:

- a) neocerebellar syndrome.
- b) tabes dorsalis.
- c) hyperthyroidism.
- d) UMNL.

(12) Tendon jerks are exaggerated in:

- a) anterior horn cell disease.
- b) dorsal root lesions.
- c) Chorea.
- d) internal capsule lesion.

(13) Golgi tendon organ:

- a) has a threshold of stimulation equal to the muscle spindle.
- b) lies in parallel with the extra-fusal muscle fibers.
- c) when stimulated reflexly cause the contraction of the muscle in the tendon of which it is situated.
- d) when stimulated , reflexly relaxes the muscle in the tendon of which it is situated.

(14) The ankle jerk is exaggerated:

- a) immediately after complete transection of spinal cord at mid-thoracic region.
- b) in Parkinsonism.
- c) in neocerebellar disease.
- d) on the right side after damage in the left internal capsule.

(15) Muscle tone is reduced in the following conditions, EXCEPT:

- a) lower motor neuron lesion.
- b) upper motor neuron lesion.
- c) cerebellar lesions.
- d) toxicity with curare-like drugs.

(16) The inverse stretch reflex:

- a) depends on a receptor which is situated in the middle of the muscle.
- b) is elicited when the muscle is subjected to excessive stretch.
- c) leads to increased force of muscle contraction.
- d) is a monosynaptic reflex.

(17) The withdrawal reflex:

- a) is monosynaptic.
- b) is polysynaptic.
- c) is more rapid than the stretch reflex.
- d) disappears in complete transection of the spinal cord below level of the lesion

(18) Motor neurons in the spinal cord may be the site of convergence of signals arising from all the following sites, EXCEPT:

- a) cerebral cortex.
- b) cerebellum.
- c) reticular formation.
- d) cells in the dorsal root ganglia.

(19) Regarding the crossed extensor reflex, all are true, EXCEPT:

- a) is an antigravity reflex.
- b) is a polysynaptic spinal reflex.
- c) has a shorter latency than the stretch reflex.
- d) never appear after spinal cord transection.

(20) Concerning the flexor withdrawal reflexes:

- a) they involve reciprocal innervation.
- b) their central delay equals 0.5 msec.
- c) it is a monosynaptic reflex.
- d) it provides an example of a deep reflex.

(21) Acute transection of the lower thoracic spinal cord results in all, EXCEPT:

- a) spinal cord shock below the section, lasting for 2-3 weeks.
- b) retention of urine with overflow.
- c) paralysis of respiratory muscles.
- d) loss of all function distal to block.

(22) In complete transection of spinal cord:

- a) sudden death occurs if the lesion is above the 5th cervical segment.
- b) paraplegia if the lesion is in the mid thoracic region.
- c) all of the above are wrong.
- d) a and b are correct.

(23) Spinal shock is associated with:

- a) mass reflex.
- b) paraplegia in flexion.
- c) paraplegia in extension.
- d) retention with overflow of urine.

(24) Damage of the semicircular canal will impair the person's ability to maintain equilibrium:

- a) when the body is at rest in the prone position.
- b) when the body is at rest in the supine position.
- c) in rapid body movements.
- d) in slow body movements

(25) Maintenance of equilibrium depends on information reaching C.N.S. from the following structures, EXCEPT:

- a) proprioceptors in muscle and joints.
- b) cochlear nerve.
- c) retina
- d) vestibular apparatus.

(26) A lesion of Broca's area produces:

- a) apraxia.
- b) motor aphasia.
- c) agraphia.
- d) word deafness.

(27) Which of the following is characteristic of events occurring at excitatory synapses?

- a) synaptic vesicles bind to postsynaptic membrane .
- b) voltage-gated potassium channels are closed .
- c) efflux of calcium from the presynaptic terminal.
- d) ligand-gated sodium channels are opened

(28) The ascending reticular formation. All are true, EXCEPT:

- a) is activated by collateral branches of sensory neurons transmitting impulses to the thalamus.
- b) sends impulses to most parts of the cerebral cortex.
- c) when stimulated electrically tends to increase alertness.
- d) it relays mainly in specific thalamic nuclei.

(29) The otolith organs are receptors for all , EXCEPT:

- a) linear acceleration.
- b) maintenance of static equilibrium.
- c) orientation of the head in space.
- d) vibration.

(30) All are true about the vestibular apparatus, EXCEPT:

- a) bony labyrinth contain 3 SCCs.
- b) helps to maintain equilibrium.
- c) is stimulated only during linear acceleration.
- d) is stimulated during linear and angular acceleration.

(31) During rotation in one plane, all are true, EXCEPT:

- a) the hair cells of the cristae are stimulated.
- b) cristae of the horizontal canals are stimulated when it bends towards the utricle.
- c) cristae of vertical canals is stimulated when it bends away from the utricle.
- d) all SCCs are stimulated.

(32) All are true about utricles and sacculi, EXCEPT:

- a) have their cells arranged in different planes.
- b) are stimulated by changing the position of the head.
- c) have cupola as their receptors.
- d) respond to angular acceleration

(33) The larger area of representation in the motor cortex includes muscles:

- a) that extend the shoulder.
- b) that flex the knee.
- c) of the trunk.
- d) of vocalization.

(34) Regarding the primary motor cortex (area 4), all are true, EXCEPT:

- a) is responsible for fine delicate movements.
- b) parts of the body are represented in an area which is directly proportional in size to the complexity of movement carried by that part.
- c) sends facilitatory impulses to gamma motor neurons.
- d) is inhibitory to muscle spindles.

(35) The cortical motor area 4 on the left side of the body:

- a) there is a map representing body parts on the right side.
- b) the body is represented in an erect (upright) pattern.
- c) controls motor activity of the left side of the body.
- d) the area representing a particular body part is directly proportional in size to the size of that part or organ.

(36) The pyramidal tract:

- a) arises only from area 4.
- b) supplies only the same side of the body.
- c) most of the fibres do not cross.
- d) responsible for fine movements.

(37) Most of fibres of the pyramidal tract have direct synapse with:

- a) anterior alpha motor neurons.
- b) gamma motor neurons.
- c) interneurons of the spinal cord.
- d) dorsal horn cells of spinal cord.

(38) The pyramidal tract originates from:

- a) all areas of cerebral cortex including parietal, frontal, temporal and occipital lobes.
- b) motor, premotor areas as well from somatic sensory cortex.
- c) area 4 only.
- d) motor area 4 and premotor area 6 only.

(39) In controlling the fine movement of the hands, the corticospinal tract terminates directly on:

- a) Renshaw cells .
- b) spinal cord interneurons .
- c) spinal alphamotor neurons .
- d) posterior horn cells .

(40) Damage which interrupt upper motor neurone pathways in the human would lead to all, EXCEPT:

- a) paralysis and loss of fine voluntary movements especially of the hand fingers.
- b) wasting of muscles.
- c) loss of abdominal reflexes.
- d) positive Babinski sign.

(41) UMN lesion is manifested by:

- a) spastic paralysis.
- b) fibrillation potential.
- c) muscle wasting.
- d) normal planter response.

(42) Babinski sign is present in all, EXCEPT:

- a) pyramidal lesion.
- b) extrapyramidal lesion.
- c) cerebellar lesion.
- d) deep sleep.

(43) Babinski sign is present in:

- a) pyramidal tract lesion.
- b) chorea.
- c) children over 5 years.
- d) lower motor neuron lesion

(44) All are true about LMN lesion, EXCEPT:

- a) loss of superficial reflexes but exaggerated deep reflexes.
- b) loss of tone of affected muscle.
- c) fasciculation of the affected muscle.
- d) wasting of the affected muscle.

(45) Babiniski sign is characterized by:

- a) plantar flexion of the great toe and fanning out of the other toes.
- b) an extensor planter response.
- c) spontaneous involuntary flexor movements of the stimulated limb.
- d) dissociated sensory loss.

(46) In lower motor neuron disease, the subject may show:

- a) wasting of muscles.
- b) rigidity of muscles.
- c) hyperactive reflexes.
- d) clonus.

(47) Babinski sign is present in:

- a) UMNL.
- b) cerebellar lesion.
- c) basal ganglia lesion.
- d) complete section of a peripheral nerve.

(48) The lower motor neuron lesion is characterized by all, EXCEPT:

- a) loss of voluntary movements.
- b) lead-pipe rigidity.
- c) fasciculation of the affected muscle.
- d) a reaction of degeneration of the affected muscles.

(49) The LMN lesion is characterized by :

- a) loss of voluntary movements but intact reflex activity.
- b) wasting of the muscle involved.
- c) normal conducting velocity in the motor neurons involved.
- d) a normal electromyogram

(50) Hemiplegia affecting the right side of the body. All are true, EXCEPT:

- a) is characterized by spasticity of muscle on the right side of the body.
- b) is usually accompanied by marked muscle wasting on the right side of the body.
- c) is associated with increased tendon reflexes on the right side of the body.
- d) can be due to bleeding in the left internal capsule.

(51) Lesion of the internal capsule results in:

- a) spastic paralysis of the contralateral side of the body.
- b) flaccid paralysis of both sides of the body.
- c) reaction of degeneration on affected side.
- d) exaggerated tendon jerk on the same side of the lesion.

(52) LMNL:

- a) may be due to poliomyelitis.
- b) is localized to the affected side.
- c) is associated with atonia.
- d) all of the above.

(53) Parkinson's disease:

- a) is caused by a lesion in the substantia nigra and globus pallidus.
- b) is caused by lesion in the posteroventral nucleus of the thalamus.
- c) is accompanied by kinetic tremors.
- d) is accompanied by clasp-knife rigidity.

(54) All of the following are features of Parkinsonism, EXCEPT:

- a) a tremor at rest.
- b) a deficiency of dopamine in the substantia nigra.
- c) a clasp-knife type of spastic rigidity.
- d) difficulty in initiating the voluntary movements.

(55) Parkinson's disease is characterized by:

- a) tremors made worse by movements.
- b) paralysis of one side of the body.
- c) increase of dopamine in the substantia nigra.
- d) lead-pipe rigidity.

- (56) Parkinsonism results in all, EXCEPT:**
- a) akinesia.
 - b) kinetic tremors.
 - c) mask face.
 - d) hypertonia.
- (57) Parkinsonism is associated with:**
- a) kinetic tremors.
 - b) adidochokinesia.
 - c) rebound phenomenon.
 - d) shuffling gate.
- (58) Chorea:**
- a) is due to lesion in the substantia nigra.
 - b) may be due to lesion in the internal capsule.
 - c) is associated with hypotonia.
 - d) is associated with exaggerated tendon jerk.
- (59) Concerning the basal ganglia function. All are true, EXCEPT:**
- a) caudate nucleus plan the sequence of patterns of movement.
 - b) caudate nucleus modifies the timing of movements.
 - c) putamen helps execution of subconscious learned pattern of movement.
 - d) basal ganglia mainly increases the muscle tone.
- (60) Which of the following doesn't occur, after spinal cord injury?**
- a) there is spinal shock immediately after injury .
 - b) visceral and deep reflexes are intact below the level of lesion .
 - c) loss of vasomotor tone below the level of lesion .
 - d) loss of voluntary movement below the level of the lesion .
- (61) Widespread damage of the basal ganglia in humans produces:**
- a) flaccid paralysis .
 - b) decreased hearing acuity .
 - c) increased capacity to perform fine movements .
 - d) increased muscle tone.

(62) The neurotransmitter used by the axons of substantia nigra neurons which project to the caudate and putamen is :

- a) acetylcholine.
- b) serotonin.
- c) dopamine .
- d) norepinephrine .

(63) Which one of the following is false regarding the cerebellum:

- a) each cerebellar hemisphere controls movements of the ipsilateral muscles.
- b) cerebellar ataxia is a sensory ataxia.
- c) cerebellar lesions cause pendular knee jerks.
- d) the intention tremor associated with cerebellar disease is worsened by movement

(64) Cerebellum is concerned with all, EXCEPT :

- a) the planning of motor movements
- b) the co-ordination of the voluntary movements.
- c) neocerebellum causes excitation of muscle spindles
- d) detection of linear acceleration.

(65) Concerning cerebellum:

- a) each cerebellar hemisphere acts on contralateral muscles.
- b) spinocerebellar tracts give rise to conscious recognition of joint position sense.
- c) paleocerebellar is facilitatory to muscle tone.
- d) neocerebellum helps planning of movements.

(66) The functions of the cerebellum includes:

- a) stop the movement at the precise intended point.
- b) prevents oscillation of movements.
- c) planning and timing function.
- d) all of the above.

(67) Regarding the cerebellum:

- a) has no role in maintenance of balance.
- b) neocerebellum inhibits muscle tone.
- c) paleocerebellum facilitates muscle tone.
- d) cerebellum has a predictive function during running.

(68) Neocerebellar syndrome in humans is characterized by all, EXCEPT:

- a) hypotonia.
- b) pendular knee jerk.
- c) disturbance of balance.
- d) static tremors.

(69) Characterized features of cerebellar disease include:

- a) hyperalgesia.
- b) nystagmus.
- c) static tremors.
- d) hypertonia.

(70) Adiadochokinesia:

- a) is failure of the patient to perform rapid coordinate movements.
- b) is due to lesion in caudate nucleus.
- c) is due to lesion in subthalamus.
- d) is due to lesion in red nucleus.

(71) The following are neurological features of cerebellar lesion, EXCEPT:

- a) abnormal finger-nose test.
- b) rebound phenomenon.
- c) dysdiadochokinesia.
- d) shuffling gait.

(72) All about kinetic tremors is true, EXCEPT:

- a) occurs during rest.
- b) is due to neocerebellar lesion.
- c) is accompanied with hypotonia.
- d) disappear during sleep.

(73) Neocerebellum syndrome is associated with all, EXCEPT:

- a) ataxia.
- b) hypertonia.
- c) inability to do complex movements.
- d) rebound phenomenon.

(74) Neocerebellum disease is associated with:

- a) shuffling gait.
- b) drunken gait.
- c) stamping gait.
- d) circumduction gait.

- (75) A cerebellar lesion is most likely associated with severe loss of equilibrium when the lesion involves:**
- a) the neocerebellum.
 - b) a small portion of the right hemisphere.
 - c) a small portion of the left hemisphere.
 - d) the flocculonodular lobe.
- (76) The following are neurological features of cerebellar lesion, EXCEPT:**
- a) dysmetria.
 - b) lead-pipe rigidity.
 - c) dysdiadochokinesia.
 - d) rebound phenomenon.
- (77) Delta waves can be obtained in EEG of a:**
- a) normal wakeful person.
 - b) during REM sleep.
 - c) person who sees a bright light.
 - d) non-REM sleep.
- (78) EEG record of relaxed subject, who closes his eyes include the following wave:**
- a) alpha.
 - b) beta.
 - c) theta.
 - d) delta.
- (79) If the subject think or open his eyes, EEG record wave will be:**
- a) alpha.
 - b) beta.
 - c) theta.
 - d) delta
- (80) When the subject open his eyes, EEG waves become:**
- a) alpha.
 - b) theta.
 - c) delta.
 - d) desynchronized.

- (81) All is true about alpha waves of EEG, EXCEPT:**
- a) disappear when a person solves a problem.
 - b) replaced by beta waves during mental activity.
 - c) has a frequency of 30 c/sec.
 - d) has an amplitude of 50 μ v.
- (82) Which of the following observation would suggest that an upper motor neuron lesion rather than a lower motor neurone lesion is present ?**
- a) marked muscle atrophy .
 - b) hyporeflexia.
 - c) plasticity .
 - d) fasciculations.
- (83) Which of the following structures is maximally sensitive to linear movement of head in the horizontal plane?**
- a) macula of saccule .
 - b) macula of utricle .
 - c) crista ampullaris of semi-circular canals .
 - d) proprioceptors in the neck .
- (84) Slow wave sleep is associated with:**
- a) increased respiratory rate.
 - b) increase muscle tone.
 - c) decrease ABP.
 - d) dream which can be remembered.
- (85) A patient who suffers from hemiplegia is able to write his thought to his family but is unable to express these ideas as spoken words is affected by:**
- a) visual agnosia.
 - b) sensory aphasia.
 - c) motor aphasia.
 - d) apraxia.
- (86) Motor apraxia:**
- a) is due to lesion in primary motor area 4.
 - b) is due to lesion in sensory area.
 - c) is accompanied with muscle paralysis.
 - d) is characterized by loss of memory of voluntary movements.

(87) During REM sleep:

- a) muscle tone throughout the body is decreased .
- b) heart rate and respiration are regular .
- c) individuals rarely awake spontaneously .
- d) is a dreamless sleep .

(88) The portion of the basal ganglia concerned with control of cognitive motor activity (memory guided) involves the following:

- a) Putamen.
- b) caudate nucleus .
- c) globus pallidus .
- d) subthalamus .

(89) Horizontal rotation of the head is likely to stimulate:

- a) macula of the utricle .
- b) crista ampullaris of the posterior semicircular canal.
- c) crista ampullaris of the anterior semicircular canal.
- d) crista ampullaris of the horizontal semicircular canal.

(90) Concerning the functions of the cerebellum the following statements are true, EXCEPT:

- a) the cerebellum co-ordinates the muscle contractions required to execute a movement.
- b) the cerebellum directly stimulates motor neurons required to make a movement.
- c) the cerebellum receives feedback from muscles that execute the movement.
- d) the cerebellum is involved in the planning of a movement.

CENTRAL NERVOUS SYSTEM

Special Senses

1. The fovea:

- a) is more vascular than the periphery of the retina
- b) is composed of cones only
- c) contains a greater proportion of ganglion cells
- d) contains a greater proportion of rods than the periphery of the retina

2. Aqueous humour:

- a) pressure is about 90 mmHg .
- b) is easily absorbed when the pupil is widely dilated .
- c) formation is dependent on carbonic anhydrase enzyme .
- d) is absorbed by the choroids plexus .

3. The primary visual cortex:

- a) is located in the temporal lobe .
- b) lies adjacent to the calcarine fissure of the occipital cortex .
- c) contains areas for form, colour and details of objects .
- d) is called V-2 .

4. The neural elements responsible for the accommodation of the lens when the eyes are focused on a near object include:

- a) the parasympathetic fibers in the trochlear nerve .
- b) the parasympathetic fibers in the oculomotor nerve .
- c) sympathetic nerve fibers in oculomotor nerve .
- d) sympathetic nerve fibers in abducent nerve .

5. The condition in which the lens of the eye becomes totally unaccommodating in elderly persons is:

- a) myopia.
- b) emmetropia .
- c) presbyopia .
- d) hypermetropia .

6. The intraocular fluid flows from the canal of schlemm into:

- a) posterior chamber .
- b) anterior chamber .
- c) aqueous veins.
- d) lens .

- 7. The intraocular fluid (aqueous humor) is secreted by the following:**
- a) iris .
 - b) cornea .
 - c) lens .
 - d) ciliary processes.
- 8. Which of the following photoreceptors respond to the longest wavelength of light?**
- a) rods .
 - b) green cones .
 - c) blue cones .
 - d) red cones .
- 9. Regarding colour vision, which of the following statements is true?**
- a) the stimulation ratio of the three types of cones allows specific colour perception .
 - b) yellow colour is perceived when green and blue cones are stimulated equally .
 - c) when there is no stimulation of red, blue or green cones, there is the sensation of seeing white .
 - d) green is perceived when only green cones are stimulated .
- 10. In response to light, the rods hyperpolarize as a result of:**
- a) an increase in sodium pump activity of the inner segment.
 - b) a decrease in Na^+ conductance into the outer segment.
 - c) an increase in potassium conductance into the outer segment.
 - d) a decrease in sodium pump activity at the inner segment.
- 11. Lesion of the optic chiasma will lead to:**
- a) homonymous hemianopia .
 - b) bitemporal hemianopia .
 - c) blindness in both eyes .
 - d) Protanopia.
- 12. Regarding light adaptation, the following is likely to occur:**
- a) pupil size increases .
 - b) vitamin A converts into retinal.
 - c) rhodopsin concentration is very high .
 - d) photopigments in rods and cones are reduced into opsin and retinal.

13. Colour blindness:

- a) is more common in women than men.
- b) is a condition linked to the Y-chromosome.
- c) results from inability to detect one of the three primary light colours red, yellow and green .
- d) no colours can be detected due to failure of all the cone systems.

14. In someone with myopia:

- a) the near point is farther than normal from the eye.
- b) near vision is affected more than distance vision.
- c) the eye tends to be longer than average from lens to retina.
- d) a convex lens can correct the refractr error.

15. In long-sightedness (hyper-metropia) :

- a) objects at the usual near point are focused behind the retina.
- b) objects at infinity can not be focused sharply on the retina.
- c) the near-point can be brought closer to the eye by the use of biconcave lens.
- d) the eye tends to be longer than average from lens to retina.

16. On entering a darkened room, the:

- a) threshold light intensity for the eye starts to rise.
- b) adaptation is slower if a long period was spent in bright light beforehand.
- c) first phase of adaptation is mainly by the rods.
- d) final phase of adaptation is mainly by the cones.

17. Constriction of the pupil increases:

- a) refractive power of the eye.
- b) spherical aberration.
- c) depth of focus.
- d) amount of light entering the eye.

18. During near vision, the following changes occur in the eye, EXCEPT:

- a) the visual axes of the two eyes converge.
- b) The depth of focus increases.
- c) The curvature of the cornea increases.
- d) Chromatic and spherical aberration is decreased.

19. When light is shone into one eye, the pupil:

- a) in that eye constricts and in the opposite pupil dilates.
- b) constricts even though its optic nerve has been cut.
- c) responds due to parasympathetic nerve activity.
- d) responds even after local application of atropine.

20. Presbyopia is caused by :

- a) weakness of the ciliary muscle.
- b) loss of lens elasticity.
- c) degeneration of the retina.
- d) clouding of the cornea.

21. Activation of transducin by light activates an enzyme that:

- a) activates adenyl cyclase.
- b) hydrolyzes cGMP.
- c) depolarizes the receptor membrane.
- d) releases calcium from intracellular stores.

22. Myopia is characterized by:

- a) very short anteroposterior diameter of eye from lens to retina.
- b) retinal image which form behind the retina.
- c) improvement of the condition with the development of presbyopia.
- d) proper correction by use of convex lens.

23. The Argyll Roberston pupil:

- a) do not constrict because the parasympathetic fibres to the eye are destroyed.
- b) do not constrict because the sympathetic fibres to the eye are destroyed.
- c) constrict during accommodation when looking at near object.
- d) constrict when light is shone on the eye.

24. The fovea of the eye:

- a) contains only rods.
- b) is situated over the exit of the optic nerve.
- c) is the region of highest visual acuity.
- d) is the thickest part of the retina.

25. Which of the following parts of the eye has the highest concentration of rods?

- a) fovea.
- b) peripheral part of retina.
- c) iris.
- d) optic disc.

26. Visual accommodation involves:

- a) increased intraocular pressure.
- b) increased tension on the lens ligaments.
- c) contraction of the ciliary muscle.
- d) dilatation of the pupil.

27. In response to light, the following changes occur in the rods, EXCEPT :

- a) decreased cGMP.
- b) activation of transduction.
- c) closure of Na^+ channels in the outer segment .
- d) increased release of synaptic transmitter.

28. Abnormal colour vision is caused by:

- a) recessive gene on the Y chromosome.
- b) dominant gene on the Y chromosome.
- c) recessive gene on the X chromosome.
- d) recessive gene on chromosome 7.

29. The following event occurs in photoreceptors during phototransduction in response to light:

- a) phosphodiesterase activity decreases .
- b) hydrolysis of cGMP increases .
- c) neurotransmitter release increases .
- d) number of open voltage-gated calcium channels increases .

30. The following are functions of the middle ear, EXCEPT:

- a) the Eustachian tube equalizes pressure on both sides of the oval window .
- b) impedance matching to transfer effectively sounds from the air filled middle ear to the fluid filled cochlea
- c) the constant contraction of the middle ear muscles, keeps the tympanic membrane optimally tense for vibration by sounds .
- d) attenuation of sound by the reflex contraction of middle ear muscles .

31. Which of the following is the middle ear ossicles that is attached to the tympanic membrane?

- a) malleus
- b) incus
- c) stapes
- d) modiolus

32. Tympanic reflex:

- a) protects the inner ear from the vibrations of high-frequency sounds .
- b) increases the rigidity of the ossicular system, thereby reducing conduction of high intensity sounds .
- c) occurs following a latent period of 4 to 8 seconds after a loud sound .
- d) can increase the intensity of low-frequency sound transmission by 30 to 40 decibels .

33. The tympanic membranes:

- a) have an area about four times that of the oval window .
- b) transmit sound in the 20-4000 Hz range with the least loss of energy .
- c) bulge outwards during descent in unpressurized airplane .
- d) allow sound waves to reach the oval and round windows at the same time .

34. Regarding the basilar membrane, the following is true:

- a) it separates the scala media from the scala vestibule.
- b) its fibers decrease in length from the base of the cochlea to the apex of the cochlea/
- c) its fibers increase in diameter from the base to the apex of the cochlea.
- d) it vibrates best at high frequency near the base of the cochlea whereas it vibrate at low frequency at the apex of the cochlea.

35. The organ of Corti is contained within:

- a) saccule.
- b) scala media .
- c) scala tympani.
- d) scala vestibule .

36. Which of the following muscles is contracted as part of the papillary light reflex?

- a) dilator pupillae muscle .
- b) constrictor pupillae muscle .
- c) ciliary muscle .
- d) medial rectus .

37. Concerning the discrimination of sound loudness, the following is true:

- a) amplitude of vibration of the basilar membrane increases with louder sounds
- b) with loud sounds hair cells excite nerve endings at a diminished rate .

- c) few outer hair cells are stimulated with loud sounds .
- d) few inner hair cells are stimulated with loud sounds .

38. Regarding transmission of the sound wave in the cochlea, the following is true statement:

- a) the footplate of the stapes moves inward against the round window, and the oval window bulges outward .
- b) the footplate of the stapes moves inward against the oval window and the round window bulges outward .
- c) the incus moves inward against the round window and the oval window bulges outward .
- d) the head of the malleus moves inward against the oval window, and the round window bulges outward .

39. In the middle ear:

- a) air pressure is normally subatmospheric.
- b) destruction of auditory ossicles abolishes hearing.
- c) the round window moves reciprocally with the oval window.
- d) paralysis of the middle ear muscles abolishes hearing.

40. The tympanic membrane:

- a) stops vibration almost immediately after the sound stops.
- b) transmits sound more effectively when the muscles of the middle ear contract.
- c) bulges outwards when the Eustachean tube is blocked.
- d) modifies the frequency of sound waves impinging on the ear.

41. The primary function of the middle ear bones is to:

- a) protect the ear from loud sounds.
- b) localize sounds
- c) amplify sounds.
- d) frequency discrimination.

42. Depolarization of the hair cells in the cochlea is caused by the influx of:

- a) Na^+ into the hair cells.
- b) K^+ into the hair cells.
- c) Ca^{++} into the hair cells.
- d) Cl^- into the hair cells.

- 43. If the patient is unable to hear low-frequency sounds, the damage to the basilar membrane is closest to:**
- a) oval window.
 - b) helicotrema.
 - c) modiolus.
 - d) spiral ganglion
- 44. The stereocilia of hair cells are embedded in the:**
- a) Reissner's membrane:
 - b) basilar membrane .
 - c) tectorial membrane .
 - d) rods of Corti.
- 45. Sour taste is elicited by :**
- a) alkaloids
 - b) amino acids
 - c) hydrogen ions
 - d) aldehydes
- 46. Bitter sensation is elicited by :**
- a) amino acids .
 - b) alkaloids .
 - c) Hydrogen ions.
 - d) aldehydes .

SPECIAL SENSES



1. This is (1)
2. Mention 3 reflexes that could be tested by this item and their centers. (6)
.....
.....
.....
3. Mention 2 causes of exaggerated tendon jerks. (2)
.....
.....
4. On the same side of exaggerated tendon jerk Babiniski sign is expected to be (**positive – negative – absent – exaggerated**) (1) in which there is (1) of big toe and (1) of other toes.
5. Mention 2 causes of positive Babiniski in normal person. (2)
.....
.....
6. Mention 2 causes of pendular knee jerk. (2)
.....
.....

1. What is this reflex?

.....
..... (1)

2. Response in normal persons is

.....
..... (1)

3. Its center is (1)

4. If there is **right** hemisection of spinal cord at T1 this reflex will be

..... (**absent – exaggerated – normal**) on the right side of the body, while on the left side it will be (**absent – exaggerated – normal**) (2)

5. In the patient , cremastic reflex will be **absent** on right side and **normal** on the left side.(2)

6. Babinski sign will be positive on the right and negative on the left. (2)

7. State of The muscles of the lower limb on the right side will by (**normal – atrophied – hypertrophied**) after few days of the lesion while on the left side will be (**normal – atrophied – hypertrophied**) (2)

8. Examining knee jerk on the right side of the same patient reveals (**normal – lost – exaggerated**) response. (1)

9. Examining biceps jerk in on the right side the same patient revels (**normal – lost – exaggerated**) response. (1)

10. Fine touch sensation on the abdomen will be (**intact – lost**) from the right side while pain and temperature from the left side will be (**intact – lost**). (2)



1. What is this reflex?

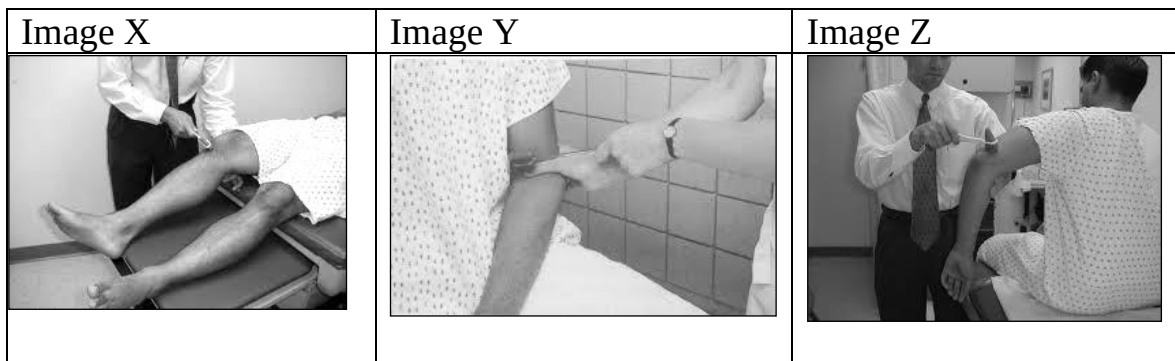
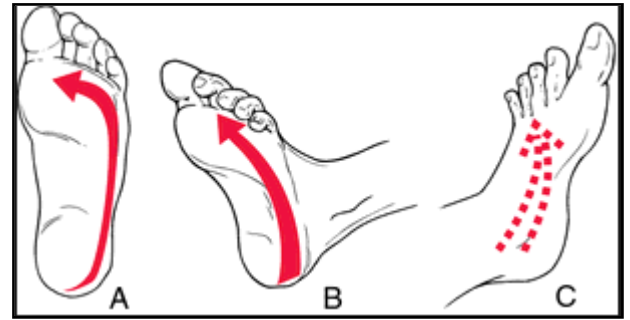
.....
 (1)

2. What is the normal response? (1)

- a) Response in image A.
- b) Response in image B.
- c) Response in image C.

3. Abnormal response occurs in (1)

4. In normal persons this reflex may be abnormal as in &
(2)



A. Name the medical tool presents in these images. (1)

.....

B. Identify the reflexes in above images(e.g. x=??). (3)

.....

.....

.....

C. Mention their spinal center (3)

.....

.....

.....

D. In case of Poliomyelitis (Viral disease destroying AHC) affecting T8,T9 and T10, describe the status (Normal / lost/ exaggerated) of each of these reflexes. (3)

.....

I. Mention the name of this instrument.(1)

.....

II. Mention one somatic sensation that could be tested using the above item. (1)

.....

III. Mention the receptor (s) responsible for this sensation. (2)

.....

.....

.....

IV. This sensation is carried on Tract (1).



Image X		Image Y	
			

1. Mention the name of the test shown in the above image. (1)

2. Image X show how to test sound conduction by to the cochlea. (1) while Image Y show how to test sound conduction byto the cochlea. (1)

3. Normal result of the above test
is (1)
4. In the above image, the tested person has perforated tympanic
membrane in his Right ear; describe the result of shown test.
..... (1)
5. If he has damage of hair cells in the organ of corti the result of the
test will be (1)
- ////////////////////////////////////
6. Corneal reflex obtained by touching the cornea by piece of cotton.
The normal response will be..... (2)
7. Injury of optic nerve of the touched eye abolishes the corneal
reflex (**true – false**) explain your answer (2)
.....
.....
8. What is meant by visual acuity of a person is 6/60? (2)
.....
.....
9. Visual acuity is poor in myopia and high in hypermetropia (true –
false) explain your answer. (2)
.....
.....
10. Hypermetropia is corrected by lenses. (1)

CENTRAL NERVOUS SYSTEM

<i>Sensory Nervous System</i>				
(1) b	(2) a	(3) d	(4) b	(5) B
(6) c	(7) b	(8) a	(9) b	(10) A
(11) a	(12) b	(13) c	(14) b	(15) A
(16) b	(17) a	(18) d	(19) c	(20) A
(21) d	(22) c	(23) c	(24) c	(25) A
(26) c	(27) a	(28) d	(29) d	(30) C
(31) b	(32) a	(33) d	(34) a	(35) C
(36) b	(37) b	(38) a	(39) b	40) C
(41) c	(42) b	(43) a	(44) b	(45) B
(46) c	(47) b	(48) c	(49) d	(50) B
(51) a	(52) c	(53) a	(54) c	(55) A
(56) c	(57) c	(58) b	(59) a	(60) B
(61) b	(62) b	(63) a	(64) c	(65) C
(66) d	(67) b	(68) d	(69) b	(70) B
(71) a	(72) b	(73) c	(74) c	(75) C
(76) c	(77) c	(78) c	(79) b	(80) C
(81) d	(82) a	(83) b	(84) b	(85) D
(86) d	(87) b	(88) d	(89) b	(90) D

<i>Motor Nervous System</i>									
(1)	b	(2)	c	(3)	c	(4)	c	(5)	A
(6)	c	(7)	a	(8)	d	(9)	d	(10)	B
(11)	a	(12)	d	(13)	d	(14)	d	(15)	B
(16)	b	(17)	b	(18)	b	(19)	c	(20)	A
(21)	c	(22)	d	(23)	d	(24)	c	(25)	B
(26)	b	(27)	d	(28)	d	(29)	d	(30)	C
(31)	d	(32)	d	(33)	d	(34)	d	(35)	A
(36)	d	(37)	c	(38)	b	(39)	b	40)	B
(41)	a	(42)	c	(43)	a	(44)	a	(45)	A
(46)	a	(47)	a	(48)	b	(49)	b	(50)	B
(51)	a	(52)	a	(53)	a	(54)	c	(55)	D
(56)	b	(57)	d	(58)	c	(59)	d	(60)	B
(61)	d	(62)	c	(63)	b	(64)	d	(65)	D
(66)	d	(67)	d	(68)	d	(69)	b	(70)	
(71)		(72)		(73)		(74)		(75)	D
(76)	b	(77)	d	(78)	a	(79)	b	(80)	D
(81)	c	(82)	c	(83)	b	(84)	c	(85)	C
(86)	d	(87)	a	(88)	b	(89)	d	(90)	B

<i>SPECIAL SENSE</i>									
(1)	b	(2)	c	(3)	b	(4)	b	(5)	C
(6)	c	(7)	d	(8)	d	(9)	a	(10)	B
(11)	b	(12)	d	(13)	c	(14)	c	(15)	A
(16)	b	(17)	c	(18)	c	(19)	c	(20)	B
(21)	b	(22)	c	(23)	c	(24)	c	(25)	B
(26)	c	(27)	c	(28)	c	(29)	b	(30)	A
(31)	a	(32)	b	(33)	b	(34)	d	(35)	B
(36)	b	(37)	a	(38)	b	(39)	c	40)	A
(41)	c	(42)	b	(43)	b	(44)	c	(45)	C
(46)	b								